1. Which of the Following Life Phenomena Involves a Chemical Change?

A. Refrigerator Cooling

B. Rust on a Cast-Iron Pot

C. Glass Shattered

D. **Alcohol volatilization**

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2. Which of the following substance names corresponds correctly to their chemical formulas?

A. Caustic soda – Na2CO3

B. Lime — Ca(OH)2

C. Soda ash – NaHCO3

D. Dichloride aluminate—$CuSO_4 \cdot 5H2O$

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3. Which of the following organic compound nomenclature is incorrect?

A.CH3CH2CH2CH3—**Butane**

B.$CH_3C{=}CCH_3$—2- **butyne**

C. $CH_3CH(CH_3)CH_2OH$—2- **Methyl-1-propanol**

D.CH3COCH3—**Acetaldehyde**

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4. Which of the following chemical terms or symbols is expressed correctly?

A.2 **One ammonium ion:**$2NH_4$

B. **Ferrous chloride:**FeCl3

C. **Helium gas:**He2

D.4 **One sulfate molecule:**$4SO_4$

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5. Among the following substances, which one is a compound?

A. liquid oxygen

B. mercury

C. drikold

D. hydrochloric acid

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6. **Among the Following Groups of Substances, Which Ones Are Allotropes of Each Other?**

A. **Ethane and ethylene**

B. **N-pentane and isopentane**

C. **Ozone and Oxygen**

D. **Gas and Deuterium**

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7. **The oxidation state of Cr in** K2Cr2O7 **is ()**

A.+3

B.+4

C.+5

D.+6

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8. **Which of the following liquids is an emulsion?**

A. **milk**

B. **syrup**

C. **muddy water**

D. **Iodine tincture**

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9. **If a substance undergoes a flame color reaction with a yellow flame, it must contain ().**

A. **Sodium element**

B. **potassium**

C. **Calcium element**

D. **Copper element**

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10. **Among the Following Substances, Which One Is a Strong Electrolyte?**

A. **ethylic acid**

B. **aqueous ammonia**

C. **sodium chloride**

D. **saccharose**

11. **Which of the Following Substances Can Conduct Electricity When Dissolved in Water but Are Themselves Non-Electrolytes?**

A. Chlorine Hydride

B. Carbon Dioxide

C. Caustic Soda

D. Copper Wire

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12. **Which of the following reactions does not belong to the four fundamental reaction types but is classified as an oxidation-reduction reaction?**

A. $2Na + 2H_2O = 2NaOH + \uparrow$ H2 units B. $CO + CuO \xrightarrow{\triangle} Cu + CO_2$

C. $Fe_2O_3 + 6HCl = 2FeCl_3 + 3H_2O$

D. $2H_2O_2 \xrightarrow{MnO_2} 2H_2O + O_2\uparrow$

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13. **Which of the following substances is commonly used as an oxidizing agent in reactions?**

A. granulated zinc

B. carbon monoxide

C. chlorine

D. hydrogen

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14. **Among the following groups of particles, which ones are isotopes of each other?**

A.H2 **and** D2

B.16o ^{1} 0 **and** O2

c.35C1 **and 37C1**

D. **Diamond and graphite**

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15. **Which of the Following Aqueous Solutions Has a PH Greater than 7 at Room Temperature?**

A.KCl

B.Na_2CO_3

C.NH_4Cl

D.$Al_2(SO4)3$

16. When purple litmus solution is added to a colorless solution, if the solution turns blue, then the solution () A. must be alkaline

B. **It is certain to be acidic**

C. **May appear neutral**

D. **must be neutral**

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17. Among the following gases, which can be collected using both the drainage method and the downward air displacement method?

A.H_2

B.CO_2

C.O_2

D.SO_2

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18. Mix 100 mL of 0.2 mol/L Na2SO4 solution with 400 mL of water to obtain a solution with a molar concentration of Na+ approximately

For ()

A.0.04 mol/L

B.0.08 mol/L

C.0.16 mol/L

D.0.20 mol/L

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19. Which of the following statements is incorrect?

A. **Dry ice can be used for artificial rainfall.**

B. **Baking soda can be used to treat excessive gastric acid secretion.**

C. **Formaldehyde can be used for soaking seafood to preserve freshness.**

D. **Nitrogen can be utilized for food packaging preservation**

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20. The position of element 34 selenium (Se) in the periodic table is () A. Period 3, Group VIA

B. **VIA family at week 4**

C. **Week 4** VIIA **family**

D. **Week 3** VIIA **family**

21. If a Bottle of Concentrated Sulfuric Acid Is Left Open, Which of the Following Statements Is Correct?

A. **decreased solution mass**

B. **decreased solute mass fraction**

C. **The mass of solvent remains constant**

D. **Decreased solution pH**

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22. Among the following groups of substances, which ones are homologues?

A.CH3CH2CH2CH3 and CH3CH(CH3)2

B.CH3CH2OH and CH3OCH3

C.C_2H_4 and C3H6

D.CH3OH and CH3CH2OH

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23. Among the following reactions, which one is a substitution reaction? A. Addition of ethylene with hydrogen

B. Reaction of benzene with liquid bromine under iron catalysis

C. Ethanol combustion

D. Acetylene causes decolorization of acidic potassium permanganate

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24. Reversible reaction H 2(9) + I2(9) → 2HI(g)**reaches equilibrium. Which of the following statements is correct? A. The reaction stops**

B. The concentrations of all substances no longer change

C. The forward and reverse reaction rates are both zero

D. The reactants are completely converted into the products.

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25. Among the Following Substances, Which One Belongs to Acidic Oxides?

A.Na_2O

B.CO

C.SO_3

D.Al_2O_3

26. Regarding the Statement About Catalysts, the Correct One Is ()

A. **The catalyst does not participate in the reaction.**

B. **Catalysts can reduce the activation energy of reactions.**

C. **Catalysts can only accelerate the reaction rate.**

D. **The catalyst maintains its properties unchanged before and after the reaction.**

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27. Among the Following Groups of Substances, the Types of Chemical Bonds Contained Are Not Entirely the Same.

A. MgO and CaCl2

B.H2O and CO2

C.Na_2SO_4 and KCl

D.NH_4 Cl and NaCl

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29. **When dilute sulfuric acid is added to a solution, a colorless and odorless gas is produced. This gas causes clear lime water to become turbid, suggesting that the original solution may contain ().**

A. SO_4^{2-}

B. CO_3^{2-}

C.Cl

D.NO_3

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30. The reaction between zinc and dilute sulfuric acid belongs to ()

A. **combination reaction**

B. **decomposition reaction**

C. **displacement reaction**

D. **double decomposition**

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31. Which of the Following Elements Are in the Same Period?

A. **Hydrogen, Lithium, Sodium**

B. **Carbon, nitrogen, oxygen**

C. **Sodium, magnesium, aluminum**

D. **Fluorine, chlorine, bromine**

32. Which of the following atomic radius comparisons is correct? () A. Na <Mg <Al <Si

B.Si>Al>Mg>Na

C.Mg>Ca>Sr>Ba

D.K>Na>Li

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33. The schematic diagram of the extranuclear electron configuration of chlorine atoms is () (figure omitted)

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34. At room temperature, dilute 10 mL of 0.1 mol/L hydrochloric acid to 100 mL. The pH of the diluted solution is approximately ().

A.1

B.2

C.3

D.4

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35. The reaction 2NH3 (9) + N2 (9) + 3H2 (9) occurs in a 2 L sealed container, with the NH3 concentration decreasing from 0.8 mol/L to

0.4 mol/L,**then** v(H2)**is ()**

A.0.05 mol·L-¹·s-1

B.0.10 mol·L-¹·s-1

C.0.15 mol·L-¹·s-1

D.0.20 mol·L-¹·s-1

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36. The reagent combination that can be used to test SO_2^- is () A. Dilute hydrochloric acid and $BaCl_2$ solution

B. **Dilute nitric acid and** AgNO3 solution

C. **Dilute sulfuric acid and** Ba(NO3)2 solution

D. **Concentrated hydrochloric acid and AgNO3 solution**

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37. After the following reactions reach equilibrium, increasing temperature or pressure will shift the equilibrium toward the forward reaction direction.

A.N2 (9) + 3H2 (9) → 2NH3 (g) (**exothermic)**

B.2NO2 (9) → N2O4 (9) (**exothermic)**

C.C(s) + H2O(g) → CO(g) + H2(9) (**endothermic)**

D.CO2 (9) + C(s) → 2CO(g) (**endothermic)**

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38. Among the following reactions, which one is an addition reaction? A. Reaction of methane with chlorine under light exposure

B. **Phenol reacts with concentrated nitric acid in the presence of concentrated sulfuric acid.**

C. **Ethylene reacts with hydrogen chloride to form chloroethane.**

D. **Ethanol reacts with oxygen under copper catalysis**

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39. Which of the following statements about organic compounds is correct? A. All organic compounds are insoluble in water.

B. **Organic compounds must contain carbon elements.**

C. **Organic compounds cannot conduct electricity.**

D. **All organic compounds are covalent compounds.**

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40. The known reaction is 2H2S + SO2 = 3S + 2H2O. If 96 g of elemental sulfur is produced, the amount of H2S consumed is ().

A.1 mol

B.2 mol

C.3 mol

D.4 mol

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41. Among the following solutions, the one with the highest molar concentration of Cl is ()

A.100 mL0.1 mol/L $MgCl_2$

B.200 mL 0.2 mol/L NaCl

C.50 mL 0.5 mol/L $AlCl_3$

D.150 mL 0.3 mol/L KCl

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42. The conditions favorable for the forward shift of the equilibrium in the reaction 2SO2 (9) + O2 (9) → 2SO3 (9) (exothermic) are () A. Increased temperature, decreased pressure

B. **Temperature reduction, pressure increase**

C. **Temperature rise, pressure increase**

D. **Temperature reduction and blood pressure lowering**

43. **Among the following substances, which one can form a brick-red precipitate with freshly prepared** Cu(OH)2 **suspension under heating conditions?**

A. **Alcohol**

B. **Acetic Acid**

C. **Amylaceum**

D. **Acetic Ether**

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44. Which of the following ion groups can coexist in large quantities in an acidic solution?

A.Na+、OH^-、Cl^-、NO_3

B. K^+、CO_3^{2-}、SO_4^{2-}、Mg^{2+}

C. NH_4^+、Al^{3+}、SO_4^{2-}、Cl^-

D. Ba^{2+}、NO_3^-、SO_4^{2-}、Na^+

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45. **A mixture of CuO and Cu was reduced with sufficient CO (18 g). After complete reaction, pure copper powder (16.4 g) was obtained. Determine the mass of CuO in the original mixture.**

A.4g

B.6g

C.8g

D.10 g

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46. Which of the Following Experimental Procedures Corresponds Correctly to the Experimental Objectives?

Operation in experiment	Experimental objective
① Add excess ammonia water dropwise to the AlCl3 solution and filter.	Preparation of Al(OH)3 precipitate
② Evaporate the FeCl3 solution to dryness	Preparation of anhydrous FeCl3
③ Add excess dilute sulfuric acid, filter, wash, and dry	Remove iron powder mixed with copper powder
④ Pass CO2 into the Na2SiO3 solution	Comparison of acidity between carbonic acid and silicic acid

A.①②

B.③④

C.①③④

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47. A white powder consists of two substances. The following experiments were conducted: ① A small amount was added to water, and the solid portion dissolved, resulting in a colorless solution. When dilute hydrochloric acid was added, bubbles were observed, and the solid dissolved completely; ② A small amount was added to dilute sulfuric acid, bubbles were produced, and solid remained after shaking. The powder may be ().

A.Na_2SO4、$BaCO_3$

B.K_2CO_3、$CaCl_2$

C.NaCl、Ag_2CO_3

D.$MgCl_2$、$CaCO_3$

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48. The overall reaction of a new type of rechargeable battery is $Pb+PbO2+2H_2SO_4=2PbSO_4+2H2O$.**Which of the following statements is correct?**

A. **During discharge, PbO2 undergoes a reduction reaction.**

B. During discharge, the negative electrode reaction is Pb2++2e-=Pb

C. **During charging, the anode is connected to the negative terminal of the power source.**

D. **During charging, the H+ concentration in the solution increases.**